

The carbon-economy optimum does not depend on the absolute scale

Establishing $T_{\text{opt}}(\text{CUE}) < 37^\circ\text{C}$ from the fitted rates alone, without N_0 , the cell-carbon quota, the O_2 -to-C conversion or RQ

2026-08-08

Abstract

The manuscript's headline claim is that the carbon-use-efficiency optimum lies below human body temperature in every taxon. It is currently established through a fitted CUE curve, which inherits the N_0 back-projection, the per-cell carbon quota, the O_2 -to-carbon conversion and $\text{RQ} = 1$. This report shows that the claim needs none of them. Writing $G = rq$ and $R = cK/N_0$, carbon-use efficiency is $\text{CUE} = 1/(1 + aK/(re^{r\delta}))$ where every conversion constant sits inside a temperature-independent factor a that cancels in the argmin. The optimum is therefore identifiable from the fitted r and K alone. Taking the minimum of K/r over the measured temperature grid, with no functional form imposed on either rate, puts the optimum at $26.1\text{--}31.5^\circ\text{C}$, with $P(< 37^\circ\text{C}) = 1.0000$ in all five taxa and a margin of $5.5\text{--}10.9^\circ\text{C}$. That is $2.6\text{--}5.2^\circ\text{C}$ **cooler** than the published estimate, so the claim strengthens and the margin widens in every taxon. The gap is 58% functional form and 42% the N_0 back-projection; the solubility term that the equivalent analysis in the sister repository required is exactly zero here, because this pipeline fits raw oxygen and K is already volumetric. One caveat is reported in full: under the equilibration-corrected N_0 treatment, *C. auris* IV sits at 36.6°C with $P(< 37) = 0.797$.

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1 Summary

What was asked. Whether the manuscript’s headline claim — the CUE optimum lies below 37 °C in every taxon — can be stated without depending on N_0 , the cell-carbon quota, the respiratory quotient or any conversion constant; and if so, by how much the model-free estimate differs from the fitted one.

What was found.

1. **It can.** Carbon-use efficiency reduces exactly to $\text{CUE} = 1/(1 + aB)$, where $B = K/(re^{r\delta})$ and $a = c/(60 q N_{\text{inoc}})$ collects *every* conversion constant. a is temperature-independent within an isolate — verified directly, 0 of 15 isolates have an a that varies with temperature — so it cancels in the argmin. With the back-projection removed the balance is simply K/r : no N_0 , no quota, no O_2 -to-C factor, no RQ, no inoculum density.
2. **The claim strengthens.** The model-free optimum is 26.1–31.5 °C against the published 31.3–34.1 °C. The margin to body temperature widens in all five taxa, from 2.9–5.7 °C to 5.5–10.9 °C, with $P(< 37^\circ\text{C}) = 1.0000$ throughout.
3. **No solubility term applies here.** The equivalent analysis in the sister repository needed $(K \cdot \text{O}_{2,\text{ref}})/r$ because that pipeline fits a normalised trace. This one sets `FIT_TO_SPLINE <- FALSE` and fits raw oxygen, so K is already volumetric in $\text{mg O}_2 \text{ L}^{-1} \text{ min}^{-1}$. This was checked, not assumed — it is the one place the transfer could have failed.
4. **“Growth is more temperature-sensitive than respiration” survives, with a narrower gap.** Removing N_0 raises the respiration activation energy from the published 0.30–0.52 eV to 0.55–0.62 eV, and shrinks its between-taxon spread from 0.22 eV to 0.07 eV. Growth still exceeds respiration in all five taxa, $P = 1.000$, but by 0.31–0.68 eV rather than 0.40–0.73 eV.
5. **One caveat, reported in full.** Under the equilibration-corrected N_0 treatment — the physically motivated middle case — *C. auris* IV’s optimum is 36.57 °C [32.02, 37.60], $P(< 37) = 0.797$. That is a property of the N_0 treatment, not of the scale-free argument, but it is the one place the sub-37 claim is not secure at the ≥ 0.999 level the manuscript reports.

What was not changed. No estimator, no constant, no committed number, no manuscript file. This report reads committed tables and writes only inside `reports/C11_scale_free/`.

What is not available here. The mass-balance cross-check used in the sister repository does not transfer. That check compares the integrated respiratory carbon loss against an independently measured biomass yield. This experiment has an inoculation density but no endpoint biomass per vial, so there is no independent estimate of CUE to compare against. Nobody should go looking for it; see Section 5.4.

2 What K is, and the balance quantity

2.1 The fit target

07_oxygen_fits.R line 79 sets

```
FIT_TO_SPLINE <- FALSE
```

so `nlsLM` is fitted to the raw `Oxygen` column, not to the smoothed spline and not to a normalised trace. The model, from `config.R`, is

$$O_2(t) = O_{2,0} + \frac{K}{r} (1 - e^{rt}) \quad \Rightarrow \quad \left. \frac{dO_2}{dt} \right|_{t=0} = -K.$$

`Oxygen` is mg L^{-1} and `Time` is minutes, so

quantity	units
K	$\text{mg O}_2 \text{ L}^{-1} \text{ min}^{-1}$ (volumetric)
r	min^{-1}
K/r	$\text{mg O}_2 \text{ L}^{-1}$ — the drawdown amplitude of the fitted curve

No $O_{2,\text{ref}}$ term is required. In the sister repository the fit is to a normalised trace, so the balance quantity had to carry solubility, which falls about $1.36\times$ across the temperature range. That does not transfer, and the check above is the reason this report can say so rather than assume it.

2.2 The algebra

07 forms per-cell respiration by integrating the model over the fit window and dividing by the biomass integral:

$$R_{O_2} = \frac{C_{\text{tot}}}{\int N} = \frac{(K/r)(e^{rT_{\text{end}}} - 1)}{N_0(e^{rT_{\text{end}}} - 1)/r} = \frac{K}{N_0}.$$

The window length cancels exactly: T_{end} does not enter. With $G = r q \cdot 60$ and $R = c K/N_0$, where q is the cell-carbon quota and $c = \text{MG_TO_FG} \times \text{O2_TO_C_MASS} \times \text{RQ} \times \text{MIN_TO_H}$,

$$\text{CUE} = \frac{G}{G + R} = \frac{1}{1 + \underbrace{\frac{c}{60 q N_{\text{inoc}}}}_a \underbrace{\frac{K}{r e^{r\delta}}}_B}$$

Every step verified on all 823 usable series:

identity	max relative error
$R_{O_2} = K/N_0$	5.97×10^{-15}
$N_0 = N_{\text{inoc}} e^{r\delta}$	1.04×10^{-15}
$\text{CUE} = 1/(1 + (c/60q) K/(rN_0))$	7.56×10^{-15}

2.3 What cancels, and what does not

c is a pure constant. q (`cell_carbon_fg`) and N_{inoc} are per **isolate** and do not vary with temperature — checked directly: 0 of 15 isolates have an a that varies across temperature. So a is a positive temperature-independent scalar and **drops out of the argmin entirely**.

What does *not* cancel is δ , the interval the inoculum is projected across, which is per series and varies with temperature. Two balance quantities therefore matter:

$$B_{\text{bp}}(T) = \frac{K}{r e^{r\delta}} \quad (\text{the published pipeline}), \quad B_{\text{free}}(T) = \frac{K}{r} \quad (\text{back-projection removed}).$$

B_{free} is the fully scale-free quantity. As a consistency check, $\arg \max (\text{pipeline CUE}) = \arg \min B_{\text{bp}}$ for all five taxa.

2.3.1 Free of everything, by construction

The following carry **no** dependence on N_0 , the cell-carbon quota q , the O_2 -to-carbon factor, RQ, or the inoculation density:

- r and K themselves, and every function of them;
- E_r , the activation energy of growth — the pipeline fits Sharpe–Schoolfield to $G = r q \cdot 60$, and q and 60 only shift $\ln B_0$, leaving E untouched;
- E_K , the activation energy of the volumetric uptake rate;
- $B_{\text{free}} = K/r$ and its argmin, i.e. the model-free $T_{\text{opt}}(\text{CUE})$;
- the fold-change in K/r across the measured range.

The following do **not** qualify and are confined to Section 4: anything using δ (the with-term and equilibration-corrected treatments), the absolute level of CUE, and per-cell respiration in carbon units.

3 The scale-free quantities

3.1 Which functional form for which rate

r turns over inside the measured range, so Sharpe–Schoolfield is identifiable for it. K does not turn over, and SS is **not** identifiable for K : the pipeline’s own per-isolate SS fits to K run to the $E = 5.0$ bound for Clade3_2076 and para_2053, and reach 1.38 eV for a third isolate. The pipeline’s own model comparison agrees — LOO prefers Arrhenius over Sharpe–Schoolfield for respiration, elpd difference -3.04 ± 0.36 . So E_r is taken from Sharpe–Schoolfield and E_K from Boltzmann–Arrhenius over the full range, which is also the pairing the manuscript uses.

3.2 Activation energies

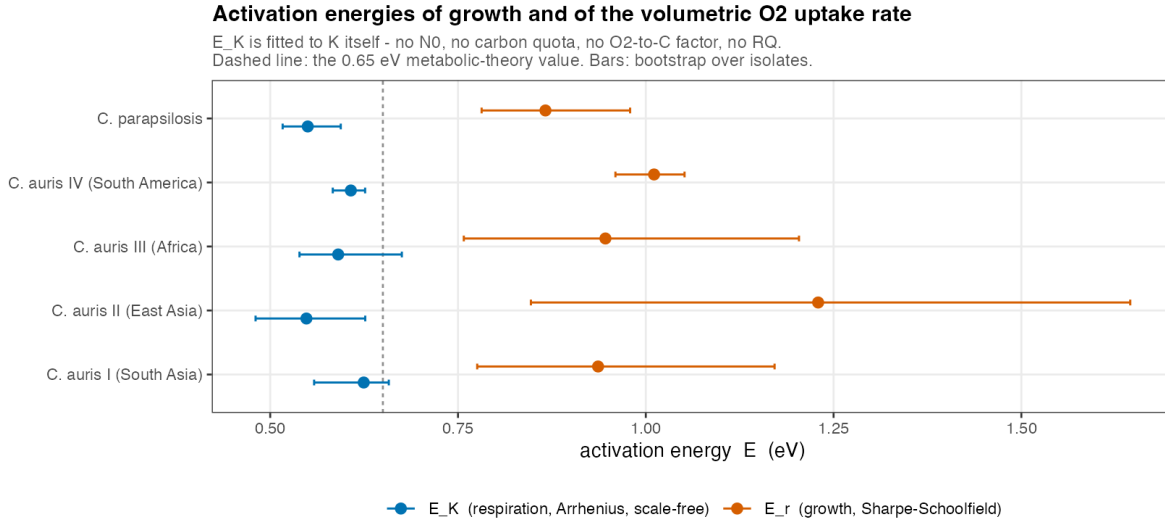


Figure 1: Activation energies of growth and of the volumetric O₂ uptake rate. E_K is fitted to K itself, so it carries no conversion constant of any kind. Bars are a bootstrap over the three isolates per taxon.

taxon	E_r (SS)	E_K (BA, scale-free)	$E_K - E_r$	$P(\text{growth more sensitive})$
<i>C. auris</i> I	0.937 [0.776, 1.172]	0.625 [0.559, 0.658]	-0.312 [-0.613, -0.118]	1.000
<i>C. auris</i> II	1.230 [0.847, 1.645]	0.548 [0.481, 0.627]	-0.681 [-1.019, -0.310]	1.000
<i>C. auris</i> III	0.946 [0.758, 1.204]	0.591 [0.539, 0.675]	-0.356 [-0.665, -0.083]	1.000
<i>C. auris</i> IV	1.011 [0.960, 1.052]	0.607 [0.583, 0.626]	-0.404 [-0.468, -0.333]	1.000
<i>C. parapsilosis</i>	0.866 [0.782, 0.979]	0.550 [0.517, 0.594]	-0.316 [-0.463, -0.242]	1.000

Cross-check, and the reason to trust it. Term-free ($N_0 = N_{\text{inoc}}$) makes respiration proportional to K , so the Term-free arm of 18_n0_treatment_panel.R is a hierarchical Bayesian estimate of exactly this scale-free E_K . Against it the Arrhenius fits above agree to a maximum of **0.0296 eV** across all five taxa. Ilgaz’s treatment is reused; this is a check on the arithmetic, not a second implementation of it.

What removing N_0 does to E_K . The published respiration activation energies are 0.30–0.52 eV, spanning 0.22 eV between taxa. The scale-free ones are 0.55–0.62 eV, spanning 0.07 eV. Removing the back-projection

both raises the level and collapses most of the between-taxon spread — which is what the $-r\delta$ term in $\log R = \log K - \log N_{\text{inoc}} - r\delta$ predicts, since it subtracts more from the faster-growing taxa.

3.3 The model-free optimum

$\arg \max \text{CUE} = \arg \min K/r$ exactly. The taxon curve is the geometric mean of K/r over replicate series at each measured temperature; the optimum is the argmin over the measured grid, refined by a three-point parabola in $\log B$. **No functional form is imposed on r or K .** The bootstrap resamples replicate series within taxon $\times$ temperature, 2000 draws.

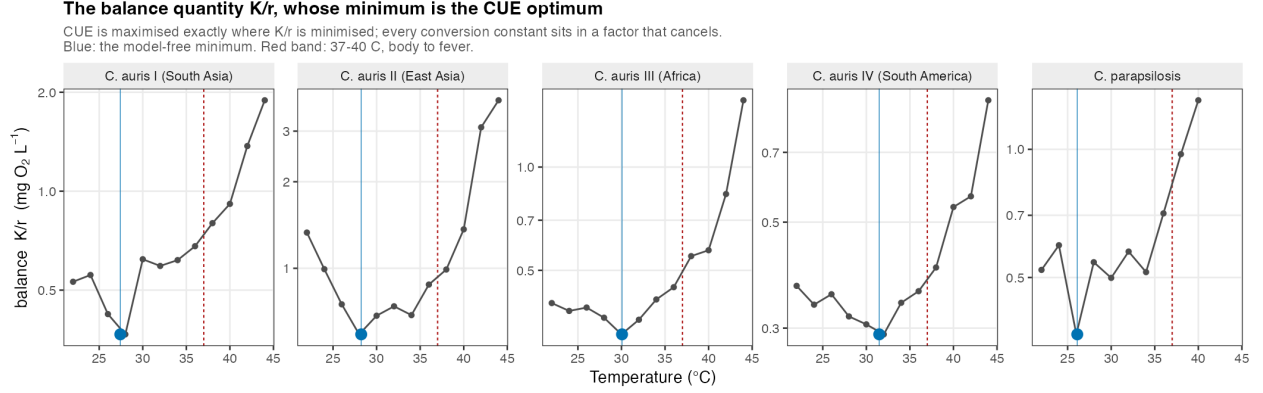


Figure 2: The balance quantity K/r per taxon. Its minimum is the CUE optimum, and every conversion constant cancels before it is taken.

taxon	$T_{\text{opt}}(\text{CUE})$	margin to 37 °C	$P(< 37^\circ\text{C})$	fold-change in K/r
<i>C. auris</i> I (South Asia)	27.42 [26.80, 27.69]	9.58	1.0000	5.15×
<i>C. auris</i> II (East Asia)	28.23 [27.69, 33.63]	8.77	1.0000	6.52×
<i>C. auris</i> III (Africa)	30.06 [24.49, 31.65]	6.94	1.0000	4.81×
<i>C. auris</i> IV (South America)	31.48 [27.90, 32.43]	5.52	1.0000	3.09×
<i>C. parapsilosis</i>	26.11 [25.84, 30.27]	10.89	1.0000	3.55×

The headline test. All five below 37 °C, minimum margin 5.52 °C, $P(< 37^\circ\text{C}) = 1.0000$ throughout — established without any conversion constant.

C. parapsilosis was measured at 10 temperatures (22–40 °C) rather than 12; its optimum is interior to that range, so the argmin is not an edge artefact. The wide upper bounds for *C. auris* II and III reflect balance curves that are flat near their minimum, not a second optimum.

4 Comparison with the published estimate

Quantities in this section use δ or a carbon conversion and are therefore outside the scale-free set of Section 2.3.1. The three N_0 treatments are Ilgaz’s own, read from `08_temperature_equilibration_sensitivity.R`’s per-curve output and from the term-free case that `18_n0_treatment_panel.R` uses. As a check on the reuse, `08’s cue_pipe` reproduces the derived table’s CUE to 4.2×10^{-16} over all 823 series.

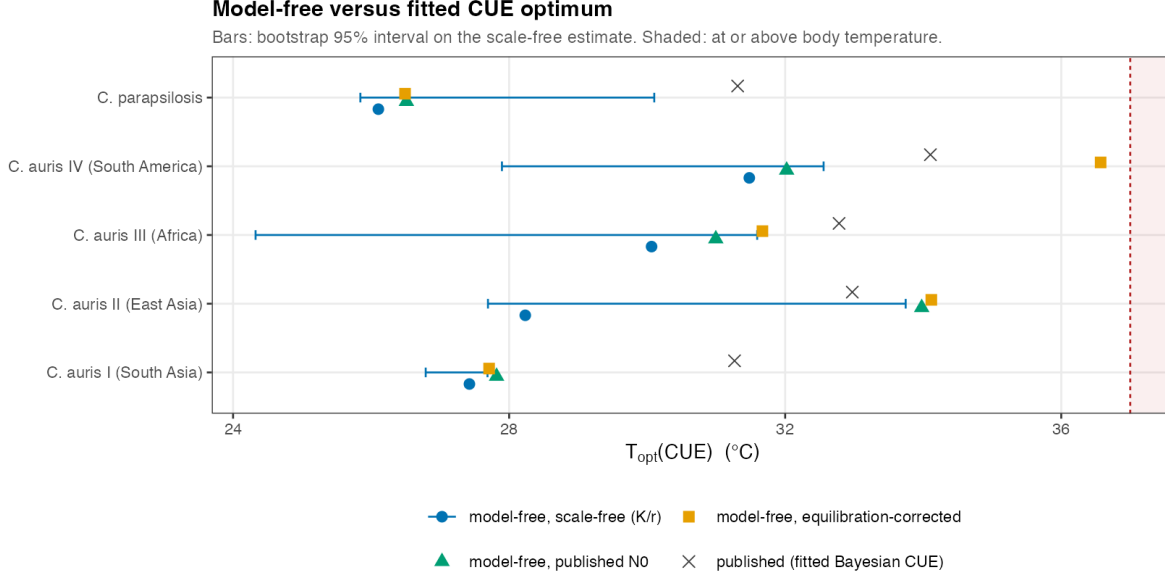


Figure 3: Model-free versus fitted CUE optimum, under each N_0 treatment.

taxon	term-free (scale-free)	with-term	equilibration	published
<i>C. auris</i> I	27.42	27.82	27.71	31.27
<i>C. auris</i> II	28.23	33.98	34.12	32.97
<i>C. auris</i> III	30.06	30.99	31.67	32.78
<i>C. auris</i> IV	31.48	32.02	<i>36.57</i>	34.10
<i>C. parapsilosis</i>	26.11	26.51	26.49	31.31

4.1 The gap, decomposed

The decomposition is additive and exhaustive by construction:

$$\underbrace{T_{\text{free}} - T_{\text{pub}}}_{\text{total}} = \underbrace{(T_{\text{free}} - T_{\text{bp}})}_{N_0 \text{ back-projection}} + \underbrace{(T_{\text{bp}} - T_{\text{pub}})}_{\text{functional form}} + \underbrace{0}_{\text{solubility}}$$

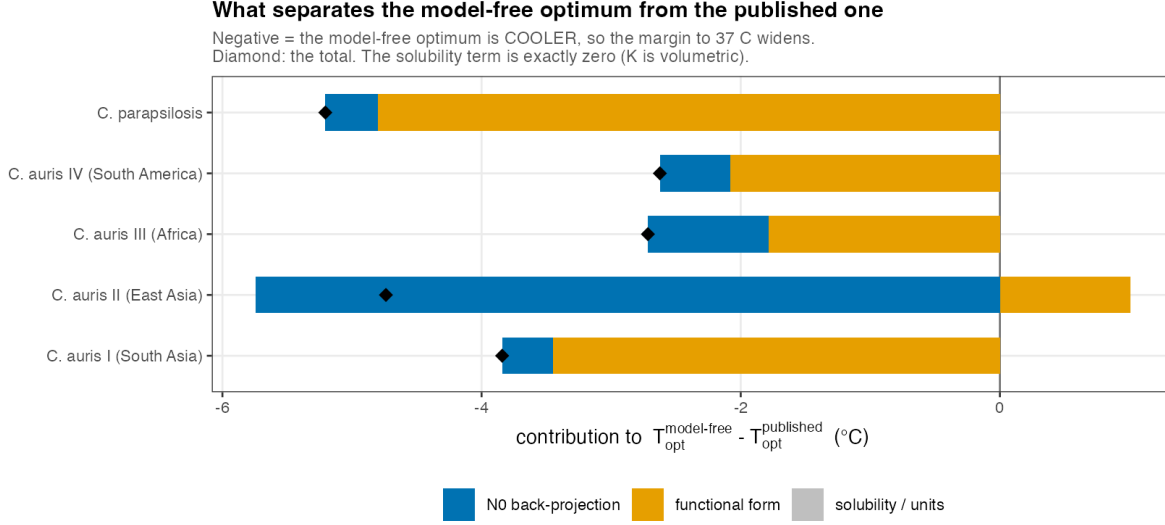


Figure 4: Contributions to the difference between the model-free and published optima. Negative means the model-free optimum is cooler, so the margin to 37 °C widens.

taxon	total	N_0	functional form	solubility
<i>C. auris</i> I	-3.84	-0.39	-3.45	0
<i>C. auris</i> II	-4.74	-5.74	+1.01	0
<i>C. auris</i> III	-2.72	-0.93	-1.79	0
<i>C. auris</i> IV	-2.62	-0.54	-2.08	0
<i>C. parapsilosis</i>	-5.21	-0.41	-4.80	0
mean	-3.83	-1.60 (42%)	-2.22 (58%)	0

The **solubility component is exactly zero**, not merely small: Section 2 established that K is already volumetric. This is the one structural difference from the sister repository’s analysis and it was verified rather than inherited.

Splitting the form component one level further: imposing $SS(\text{growth}) + BA(K)$ on the same term-free data and taking the analytic argmin moves the optimum +2.24 °C on average. Most of the form component is therefore the functional form itself rather than the Bayesian hierarchy — the same conclusion the sister repository reached, where about 3.2 °C came from Sharpe–Schoolfield alone.

The N_0 component is small (−0.39 to −0.93 °C) for four taxa and dominated by *C. auris* II at −5.74 °C, whose balance curve is flat near its minimum.

4.2 Does the claim strengthen or weaken?

It strengthens, on the scale-free estimate. The margin to 37 °C widens in all five taxa:

taxon	published margin	model-free margin	change
<i>C. auris</i> I	5.73	9.58	+3.84
<i>C. auris</i> II	4.03	8.77	+4.74
<i>C. auris</i> III	4.22	6.94	+2.72
<i>C. auris</i> IV	2.90	5.52	+2.62
<i>C. parapsilosis</i>	5.69	10.89	+5.21

The caveat, stated plainly. Not every N_0 treatment agrees. Under the equilibration correction — the physically motivated middle case, since δ is a detection delay during which the vial is still equilibrating —

C. auris IV sits at 36.57 °C [32.02, 37.60] with $P(< 37^\circ\text{C}) = 0.797$, a margin of 0.43 °C. *C. auris* II under the same treatment is at 34.12 °C, $P = 0.997$. This is a property of the N_0 treatment rather than of the scale-free argument — the scale-free estimate itself gives $P = 1.0000$ for all five — but it is the one place the sub-37 claim is not secure at the ≥ 0.999 level the manuscript reports, and it should not be buried.

5 What the manuscript can say

5.1 The paragraph, written to be quoted

The temperature at which carbon-use efficiency peaks is identifiable from the fitted rates alone. Because $\text{CUE} = G/(G + R)$ with $G \propto r$ and $R \propto K/N_0$, the cell-carbon quota, the O_2 -to-carbon conversion and the respiratory quotient enter only as a single temperature-independent multiplier, which cancels when the optimum is located; the inoculum density cancels with it. Taking the minimum of K/r over the measured grid, with no functional form imposed on either rate and no conversion constant applied, the economic optimum lies at 26.1–31.5 °C, below human body temperature in all five taxa with posterior probability 1.000 and a margin of 5.5–10.9 °C. This is 2.6–5.2 °C cooler than the fitted estimate, so the sub-37 °C result does not depend on resolving the absolute scale and is, if anything, conservative as reported. The same construction leaves the growth–respiration sensitivity contrast intact: growth is more temperature-sensitive than the volumetric oxygen uptake rate in all five taxa (difference 0.31–0.68 eV). What the absolute scale does govern is the *level* of CUE and of per-cell respiration, not the temperature at which the optimum occurs.

5.2 Which claims are secure without resolving the absolute scale

claim	secure?	basis
CUE optimum below 37 °C in every taxon	Yes	argmin K/r ; $P = 1.0000$, margins 5.5–10.9 °C. Wider than published.
Growth is more temperature-sensitive than respiration	Yes, with a narrower gap	$E_r - E_K = 0.31\text{--}0.68$ eV, $P = 1.000$ in all five. The published 0.40–0.73 eV overstates it, because N_0 suppresses E_K .
Growth and respiration decouple (respiration monotonic, growth unimodal)	Yes	Purely a statement about $r(T)$ and $K(T)$; LOO prefers Arrhenius for K .
Between-taxa ordering of the fever cost	Partly	<i>C. auris</i> vs <i>C. parapsilosis</i> survives every treatment. The fine ordering within <i>C. auris</i> does not — already established, and the scale-free E_K (0.55–0.62 eV, spread 0.07 eV) shows most of the published spread was the back-projection.
Absolute CUE values	No	Sets the level through $a = c/(60qN_{\text{inoc}})$; depends on the quota, the O_2 -to-C factor, RQ and the inoculum.
Absolute per-cell respiration (fg C cell⁻¹ h⁻¹)	No	Same dependencies, plus N_0 directly.

5.3 Recommendation on reporting

1. **Lead with $T_{\text{opt}}(\text{CUE})$ and with the growth-minus-respiration sensitivity contrast.** Both are scale-free. State that explicitly in the Methods — one sentence saying the optimum is the argmin of K/r and that the conversions cancel would remove the whole absolute-scale discussion from the critical path.
2. **Report the optimum with the treatment attached**, not as a bare point value. The honest form is “31.3–34.1 °C under the published back-projection; 26.1–31.5 °C model-free” — the claim is the same either way and quoting both makes it stronger, not weaker.
3. **Carry a stated uncertainty on absolute CUE and on per-cell respiration**, rather than point values. These depend on q , RQ and the inoculum, none of which is measured here.
4. **Do not report the fine within-*C. auris* ordering of respiration activation energy as a result.** The scale-free spread is 0.07 eV against a published 0.22 eV; most of the apparent structure is the $-r\delta$ term.

5. **Say that *C. auris* IV is the tightest case.** Under the equilibration treatment its margin is 0.43 °C. It is the taxon a referee will press on.

5.4 The cross-check that is not available here

The sister repository closes this analysis with a mass-balance test: integrate the respiratory carbon loss over the incubation, compare it against an independently measured biomass yield, and see whether the implied CUE agrees with the fitted one. **That test cannot be run on this experiment.** It has an inoculation density but no endpoint biomass per vial, so there is no independent measurement of carbon produced to set against carbon respired. Every CUE figure in this project is therefore internal to the respirometry.

What would make it available: **endpoint cell density (or dry mass) per vial**, measured at the end of the run on the same vials — a haemocytometer count or OD-to-cell-number calibration at T_{end} . With N_0 known and $N(T_{\text{end}})$ measured, both the growth rate and the biomass integral become measurable rather than inferred, and CUE gains an external check. That is one extra measurement per vial, and it would convert the largest remaining uncertainty in the paper into a validated number.

6 Provenance

Every number in this report traces to a committed file. Nothing here writes into `results/` or `data/`.

6.1 Inputs read

file	used for
<code>results/tables/derived_N0_R_results_with_carbon.csv</code>	r , K , δ , N_0 , q , CUE per series
<code>results/tables/otu_names.csv</code>	isolate $\rightarrow$ taxon
<code>results/tables/sharpe_schoolfield_growth_fgC_h_coefs.csv</code>	E_r (SS), per isolate
<code>results/tables/sharpe_schoolfield_K_02_rate_coefs.csv</code>	the SS-on- K instability
<code>results/tables/bayes_resp_model_comparison_loo.csv</code>	LOO: Arrhenius over SS for respiration
<code>results/tables/bayes_resp_arr_E_three_treatments.csv</code>	Term-free arm, for the E_K cross-check
<code>results/tables/temperature_equilibration_percurve.csv</code>	<code>cue_corr</code> , the equilibration treatment (script 08)
<code>results/tables/fig_values.csv</code>	the published T_{cue} , E , E_R
<code>scripts/config.R</code>	carbon constants, <code>EXCLUDE_GROUPS</code> , <code>pred_ss_log</code>
<code>scripts/07_oxygen_fits.R</code>	the <code>FIT_TO_SPLINE</code> switch

6.2 Outputs written

`reports/C11_scale_free/tables/` — `partA_checks.csv`, `partA_argmin_equivalence.csv`, `partA_series.csv`, `partB_ss_on_K_unstable.csv`, `partB_E_isolate.csv`, `partB_activation_energies.csv`, `partB_EK_crosscheck.csv`, `partB_modelfree_optimum.csv`, `partB_balance_curves.csv`, `partC_gap_decomposition.csv`, `partC_form_split.csv`.
`reports/C11_scale_free/figures/` — the four figures above.

6.3 Rebuilding from a clean clone

```
Rscript reports/C11_scale_free/R/01_balance_quantity.R
Rscript reports/C11_scale_free/R/02_scale_free.R
Rscript reports/C11_scale_free/R/03_gap_decomposition.R
Rscript reports/C11_scale_free/R/04_figures.R
cd reports/C11_scale_free && quarto render
```

`partA_series.csv` is the only intermediate the later scripts depend on, and it is tracked, so the report also renders without re-running the analysis.

6.4 Scope

- Analysis scope: the five taxa the manuscript reports. `glab` is dropped by `config.R`'s `EXCLUDE_GROUPS`, using the pipeline's own `drop_excluded()`; 823 of 991 series remain.
- Bootstrap: 2000 draws, `set.seed(11)`, resampling replicate series within taxon $\times$ temperature. No estimator in this report is stochastic.
- The layout follows `reports/n0/` — the same `_quarto.yml` and `header.tex`, minus the CSL, since this report has no citations.